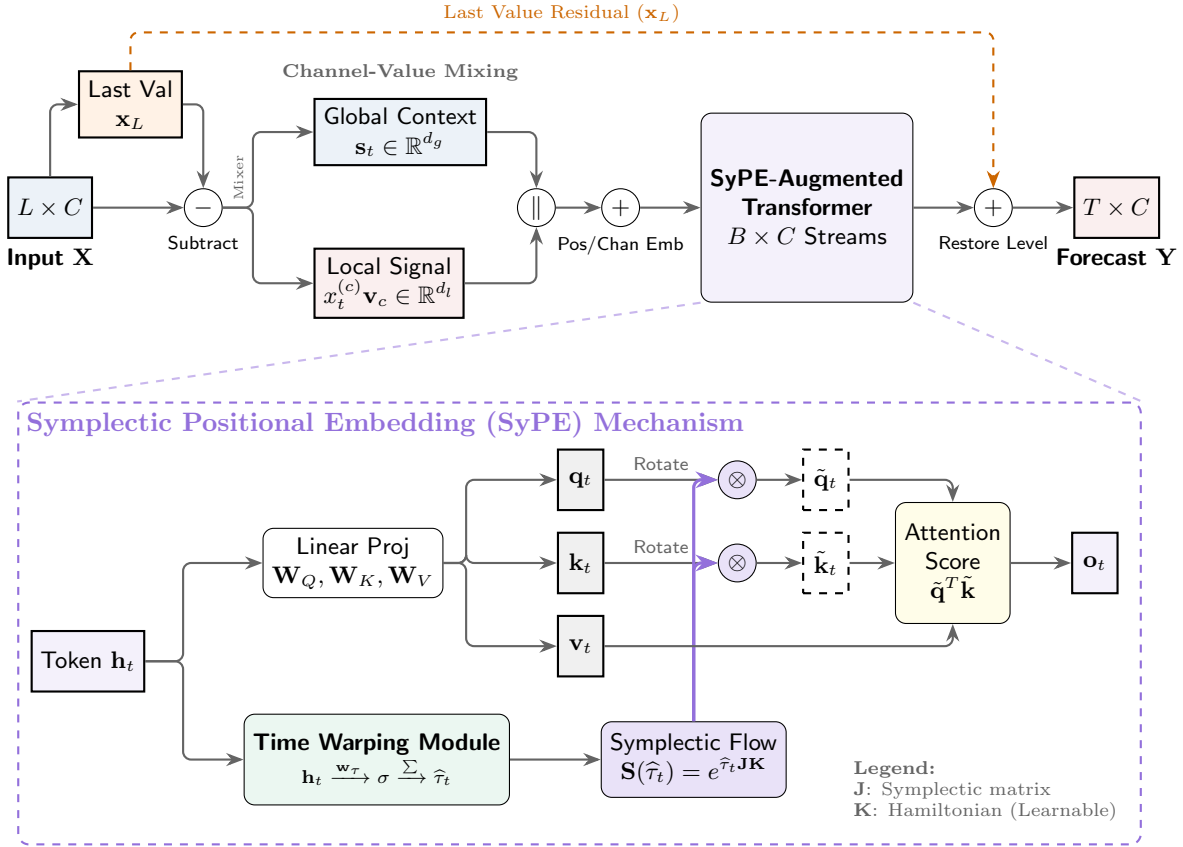
Last Value Residual ($\mathbf{x}_L$)
Channel-Value Mixing
Last Val
$\mathbf{x}_L$
$L \times C$
Input $\mathbf{X}$
Subtract
Mixer
Global Context
$\mathbf{s}_t \in \mathbb{R}^{d_g}$
Local Signal
$x_t^{(c)}\mathbf{v}_c \in \mathbb{R}^{d_l}$
Pos/Chan Emb
SyPE-Augmented Transformer
$B \times C$ Streams
Restore Level
$T \times C$
Forecast $\mathbf{Y}$
Symplectic Positional Embedding (SyPE) Mechanism
Token $\mathbf{h}_t$
Linear Proj
$\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$
$\mathbf{q}_t$
$\mathbf{k}_t$
$\mathbf{v}_t$
Rotate
Rotate
$\tilde{\mathbf{q}}_t$
$\tilde{\mathbf{k}}_t$
Attention Score
$\tilde{\mathbf{q}}^T\tilde{\mathbf{k}}$
$\mathbf{o}_t$
Time Warping Module
$\mathbf{h}_t \xrightarrow{\mathbf{w}_\tau} \sigma \xrightarrow{\sum} \widehat{\tau}_t$
Symplectic Flow
$\mathbf{S}(\widehat{\tau}_t) = e^{\widehat{\tau}_t \mathbf{JK}}$
Legend:
J: Symplectic matrix
K: Hamiltonian (Learnable)